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Unsupervised data imputation with multiple importance sampling variational autoencoders

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Recently, deep latent variable models have made significant progress in dealing with missing data problems, benefiting from their ability to capture intricate and non-linear relationships within the data. In this work, we further investigate the potential of Variational Autoencoders (VAEs) in addressing the uncertainty associated with missing data via a multiple importance sampling strategy. We propose a Missing data Multiple Importance Sampling Variational Auto-Encoder (MMISVAE) method to effectively model incomplete data. Our approach consists of a learning step and an imputation step. During the learning step, the mixture components are represented by multiple separate encoder networks, which are later combined through simple averaging to enhance the latent representation capabilities of the VAEs when dealing with incomplete data. The statistical model and variational distributions are iteratively updated by maximizing the Multiple Importance Sampling Evidence Lower Bound (MISELBO) on the joint log-likelihood. In the imputation step, missing data is estimated using conditional expectation through multiple importance resampling. We propose an efficient imputation algorithm that broadens the scope of Missing data Importance Weighted Auto-Encoder (MIWAE) by incorporating multiple proposal probability distributions and the resampling schema. One notable characteristic of our method is the complete unsupervised nature of both the learning and imputation processes. Through comprehensive experimental analysis, we present evidence of the effectiveness of our method in improving the imputation accuracy of incomplete data when compared to current state-of-the-art VAEs-based methods.

Keywords Missing data, Variational autoencoders, Multiple importance sampling, Resampling

Missing data is a ubiquitous problem in modern science and engineering. Most of statistical and machine learning tasks involve inferring missing or latent variables given observed values[1]. Incomplete data collection, sensor failures, and object occlusion are common causes of missing data[2]. Properly estimating missing data is essential for comprehensive data analysis, the presence of missing data can introduce significant biases, leading to errors in data processing and analysis, and decreasing statistical efficiency. In general, there are three categories of missing data: missing completely at random (MCAR), missing at random (MAR), and not missing at random (NMAR)[3]. In the case of MCAR, the missingness is independent of both the observed and missing variables. MAR occurs when the pattern of missing data depends solely on the observed variables. A distinctive characteristic of NMAR is that missing data depends not only on the observed variables but also on the missing values themselves. Despite substantial advancements in missing data imputation over the last few decades, accurately imputing missing data remains challenging due to the increasing scale, diversification, and complexity of data types.

Recently, deep latent variable models (DLVM) have emerged as a favored approach for modeling real-world data[4]. It effectively integrate the approximation capabilities of deep learning with the statistical foundations of generative models. Compared to traditional statistical models, DLVMs can capture complex patterns and dependencies in high-dimensional data by learning hierarchical representations in an unsupervised manner. At the core of DLVMs is the idea of incorporating latent variables, which are unobserved variables that capture underlying structure in the data. These latent variables are typically represented by lower-dimensional spaces and are used to model intricate patterns and generate more realistic data samples. However, current methods heavily rely on complete datasets and necessitate a substantial amount of fully-observed data. In many practical applications, data often exhibits varying degrees of missing values. Fortunately, statistical estimate of incomplete data shares conceptual similarities with the estimation of latent-variable models (LVM), and much work has

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been focused on applying DLVM to the problem of missing data, e.g., Variational Autoencoders (VAEs)[4,5], Generative Adversarial Networks (GANs)[6,7], and Stacked Autoencoders (SAEs)[8], etc. Besides, recent work, e.g.[7,9,10], have made significant strides in utilizing DLVM to address the missing data problem. DLVM models employ latent variables to capture the underlying structure and hidden information present in the data, enabling them to capture data variability and unobserved features effectively. With their flexible and robust modeling capabilities, DLVM models are well-suited for diverse missing patterns and various types of data, including high-dimensional[8], sequential[10], and image data[11].

Currently, VAEs-based DLVM, represented by missing data importance-weighted autoencoder (MIWAE), have achieved the state-of-the-art imputation performance. However, current VAEs-based methods exhibit limited empirical coverage for missing data, resulting in underestimation and overconfident imputations, particularly for more extreme missing data values[12]. To overcome this, we focus on enhancing the performance of VAEs in addressing statistical challenges within the context of missing data problems. One of the core problems of VAEs for missing data is find a good approximation to compute difficult-to-compute probability densities. DLVM-based methods are typically trained using approximate maximum likelihood techniques that optimize lower bounds of the log-likelihood function. This is typically by maximizing the standard evident lower bound (ELBO) or the importance weighted ELBO (IWELBO)[13] to find a suitable solution. One current research direction within DLVM is to reduce the gap between approximate and true posterior distributions, thus formulating tighter bounds. This approach involves seeking a lower bound function that effectively approximates the true distribution and utilizes variational parameter optimization during the maximization process. The application of this idea stems from the concept of variational inference[14] and serves to approximate the posterior distribution of latent variables.

In the most recent works, multiple importance sampling ELBO (MISELBO)[15,16] was proposed, which offers a tighter bound compared to the average of standard ELBOs, as evidenced by empirical demonstrations showing tighter than the average of IWELBOs. MISELBO use the mixture components to jointly adapt to maximize the ELBO. Building upon this work, we integrate MISELBO into VAEs for obtaining more flexible variational approximations to handle missing data. In summary, the contribution of this paper are as follows:

1. We improve the recently proposed MISELBO, originally designed for complete datasets, to estimate the parameters of statistical models from incomplete data. Specifically, we introduce Missing data Multiple Importance Sampling Variational Autoencoders (MMISVAE) with a target objective function that provides a tighter lower bound on the joint log-likelihood.
2. We propose a novel multiple importance resampling methods for estimating conditional distributions of the missing variables, resulting in more accurate imputation via conditional expectation and multiple imputation. The proposed imputation method is an extension of the state-of-the-art MIWAE approach.
3. We perform experiments on widely used binary image datasets, and the results illustrate the superiority of our approach over the current state-of-the-art methods, particularly in terms of image generation quality.The paper is organized as follows: "Related work" section summarizes the related work, "Background" section introduces the notation and provides the necessary background, and "Methods" section presents the proposed learning methods for training incomplete data. "Imputation" section presents the proposed multiple importance sampling imputation method. "Experiments" section provides a detailed analysis of the experiments. "Further discussions" section summarize the importance-weighted sampling methods and their connection to our recent work. The conclusion is presented in "Conclusions" section.

Related work

In recent years, a major research in DLVM has been on how to approximate the intractable maximum likelihood estimate effectively and accurately. Generative Adversarial Networks (GANs), diffusion models[17], and Variational Autoencoders (VAEs) are probably the most popular approaches.

The GANs learn a implicit generative models based on game theory, and have been made significant progress in missing data imputation. GAIN[7] combines an imputation generator and a critic network to learn the underlying data distribution and simultaneously impute missing values. The generator network generates synthetic samples that are refined by the critic network, providing a refined imputation of the missing data. MISGAN[18] combines the power of GANs with a specialized discriminator network to effectively impute missing values. The generator network learns to generate synthetic data samples, which are then refined by the discriminator network to distinguish them from the real data. Generative adversarial multiple imputation network (GAMIN)[19] combines the effectiveness of generative adversarial networks (GANs) with multiple imputation. However, the GANs based imputation methods are often need for sufficient training data, and do not inherently model uncertainty in their generated samples. Furthermore, these methods are only applicable to the Missing Completely At Random (MCAR) assumption.

Diffusion models offer a powerful framework for modeling the underlying distribution of data, including the missing values. The CSDI[10] method combines conditional scoring and diffusion models to impute time series data by learning the conditional probability distribution of the time series. DiffImpute[20] utilizes the denoising diffusion probabilistic model to estimate and restore the missing data. DiffFlowImpute[21] using diffusion and flow-based gradient-boosted trees for generating and imputing tabular data. Diffusion models can adapt to various types of missing data, including Missing At Random (MAR) and Missing Not At Random (MNAR). By modeling the evolution of data over time, diffusion models capture dependencies and latent structures among data points.

After the seminal works of the proposed VAEs[4], it has emerged as the prominent approach for imputing missing data and have gained significant attention within the research community, and has been successfully applied to many practical applications, e.g., simulated milling circuit[22], traffic estimation[23], Genomic data imputation[24] and

multimodal imputation and embedding[25]. Pereira[5] summarizes the technical trends, applications, outcomes, and provides a comprehensive evaluation of the performance of autoencoders in missing data imputation. Further, recent research efforts on improving VAEs for better missing data estimation. In Ref.[2], deep subspace learning-based VAEs was proposed for medical image imputation. Nazabal[26] made further improvements to VAEs, enabling it to handle heterogeneous (mixed continuous and discrete) missing data. Ma[27] proposed a method for handling real-valued data with heterogeneous margins and effectively learning their heterogeneous marginal distributions even in the case of partially observed values.

The most recent advances of VAEs for missing data problems are dedicated to the statistical challenges. Among these works, the most well-known method probably is MIWAE[9], which is based on the importance-weighted autoencoder (IWAE), aims to maximize a tight lower bound on joint likelihood of the incomplete data. It employs a deep generative model to learn the underlying structure and patterns of the incomplete data and uses this knowledge to impute the missing values. By leveraging deep learning techniques, MIWAE provides an effective solution for modeling complex data sets with missing values and generates reliable imputations. Since MIWAEs are valid only under MAR assumption, recent work[28,29] extends it to the case of Not Missing At Random (NMAR). As in the most recent work of Ref.[30] discussed, the single latent learning capacity is often limited and cannot capture the true posterior distribution of complex data. This entails that finding a single good proposal pdf, is not always possible. A mixture variational family method can alleviate the increase in posterior complexity caused by missing data and improve the estimation accuracy of Variational Autoencoders (VAEs) when the training data is incomplete.

In fact, Mixture Variational Autoencoders (Mixture VAEs)[31] have been proposed as a more flexible posterior approximations by incorporating mixture learning and variational objectives outside the VAEs literature. Shi[32] combines variational autoencoders and expert mixture models to establish a flexible and adaptable deep generative model for multi-modal data. In Ref.[15], the training and inference processes of the VAEs are improved by introducing Multiple Importance Sampling (MIS) and deep ensembles of variational approximations. In Ref.[16], the benefits of adding mixture components in variational autoencoders are discussed. The main idea is to explore the benefits of incorporating mixture components into VAEs. By introducing mixture models in the latent space, the modeling capability for the data distribution can be improved. The mixture components offer a more flexible model fitting, allowing VAEs to better capture the complex structures and diversities present in the data. However, since mixture distributions often involve non-differentiable categorical distributions, the reparameterization trick in Variational Autoencoders (VAEs) becomes infeasible. Most current mixture VAEs methods rely on approximate estimations derived from complete datasets. However, when handling missing data, mixture distributions typically necessitate iterative algorithms, such as the Expectation-Maximization (EM) algorithm, for approximate solutions. This requirement significantly increases the model complexity of mixture VAEs with missing data. Recent work[30] has explored the application of mixture VAEs to missing data, employing finite mixture variational distributions and utilizing approximate yet computationally efficient conditional sampling methods to generate imputations.

Mixture VAEs can be viewed as an ensemble strategy. Indeed, ensemble methods have demonstrated effectiveness in achieving state-of-the-art performance for missing data imputation. This includes the traditional statistical learning approaches such as MICE[21] and MissForest[33], as well as DLVM-based methods including GAMIN[19] and MIWAE[9]. The ensemble is typically accomplished through either multiple imputation or multiple importance sampling. Multiple Imputation by Chained Equations (MICE)[21] transforms the problem into a series of regression or classification tasks and estimates missing values by iteratively imputing them. MICE generates multiple imputed datasets to improve imputation accuracy. MissForest[33] use the random forest (ensemble of decision trees) to estimate the missing values based on observed data, which offers a flexible and robust approach that can handle various data types and missing patterns. The previously discussed mixture models[15,16,30–32] all utilize a ensemble strategy referred to as multiple importance sampling (MIS)[34–36] to estimate expectations or integrals when drawing samples from different probability distributions. By combining samples from multiple distributions, MIS improves the efficiency and accuracy of the estimates compared to using a single importance sampling distribution.

Background

Notations

Let $X = \{x^1, \ldots, x^n\} \in \mathrm{R}^{n\times d}$ denote a set of observed data, and $Z = \{z^1, \ldots, z^n\} \in \mathrm{R}^{n\times p}$ represent a set of latent variables. The superscript represents different observed samples, and the subscript represents the values of sample features. The mask vector m is defined as:

$$m_j = \begin{cases} 1 & \text{if } x_j \text{ is observed} \\ 0 & \text{if } x_j \text{ is missing.} \end{cases}$$

The set of observed elements indices of x is denoted by Ω, and $\Omega^{\perp}$ denotes the set of indices of missing elements.

Variational autoencoders (VAEs)

The inference problem is to calculate the true posterior of the latent variables given the observations, denoted as $p_{\boldsymbol{\theta}}(z \mid x)$ with parameters $\boldsymbol{\theta}$. However, computing such conditional density is often intractable. VAEs are utilized to address these issues, employing an encoder and decoder architecture. The encoder, also referred to as the inference model, employ an approximate variational posterior probability $q_{\phi}(z \mid x)$ parameters by ϕ. The variational parameters ϕ is optimized so that:

$$q_\phi(z \mid x) \approx p_\theta(z \mid x).$$

The smaller the gap between the approximate distribution $q_\phi(z \mid x)$ and the true posterior distribution $p_{\boldsymbol{\theta}}(z \mid x)$ in terms of the KL divergence, the better the approximation of $q_\phi(z \mid x)$ becomes. The decoder, also referred to as the generative model, use the latent vectors z to reconstruct the original data $\hat{x}$ via the generative model $p_\theta(x \mid z)$. In order to estimate the log-likelihood and posterior distributions, consider the following log joint probability:

$$\begin{aligned}
\log p(\mathrm{X}) &\overset{(*)}{\geq} \sum_{i=1}^{n} \mathbb{E}_{z^i \sim q_\phi(z \mid x^i)} \left[\log \frac{p\left(x^i, z^i\right)}{q_\phi\left(z^i \mid x^i\right)} \right] \\
&= \sum_{i=1}^{n} \mathbb{E}_{z^i \sim q_\phi(z \mid x^i)} \left[\log \frac{p_\theta\left(x^i \mid z^i\right) p\left(z^i\right)}{q_\phi\left(z^i \mid x^i\right)} \right] \\
&= \sum_{i=1}^{n} \mathbb{E}_{z^i \sim q_\phi(z \mid x^i)} \left[\log p_\theta\left(x^i \mid z^i\right) \right] + \mathbb{E}_{z^i \sim q_\phi(z \mid x^i)} \left[\log \frac{p\left(z^i\right)}{q_\phi\left(z^i \mid x^i\right)} \right] \\
&= \underbrace{\sum_{i=1}^{n} \mathbb{E}_{z^i \sim q_\phi(z \mid x^i)} \left[\log p_\theta\left(x^i \mid z^i\right) \right]}_{\text{reconstruction term}} - \underbrace{\sum_{i=1}^{n} \mathscr{D}_{\mathrm{KL}} \left(q_\phi\left(z^i \mid x^i\right) \| p\left(z^i\right) \right)}_{\text{prior matching term}},
\end{aligned}$$

where we used Jensen's inequality in (*), and the equality is satisfied only when the KL divergence (prior matching term) is 0 . In variational inference, the right-hand side corresponds to the Evidence Lower Bound (ELBO), which is often optimized in lieu of the log joint probability. By maximizing the ELBO, the goal is to approximate the elusive true posterior distribution through a computationally tractable variational distribution. The ELBO consists of two components: the reconstruction term (reconstruction loss) and the prior matching term (KL divergence between the encoder's distribution and the prior distribution). Maximizing the ELBO ensures that the model acquires meaningful latent representations while accurately reconstructing the data. To optimize the ELBO, VAEs often assume the following conditions:

- *Prior* As in regular factor analysis, the prior distribution of the latent variable z is often an isotropic Gaussian:

$$p(z) = \mathscr{N}(z; 0, \mathrm{I}).$$

- *Encoder* The inference models utilize a straightforward, factorized Gaussian distribution, represented by mean $\boldsymbol{\mu}_\phi$ and covariance σ_ϕ. This distribution assumes diagonal covariance and is parameterized by a neural network taking x as input, and can be represented as:

$$q_\phi(z \mid x) = \mathscr{N}\left(z; \mu_\phi(x), \sigma_\phi^2(x)\mathrm{I}\right).$$

- *Decoder* The distribution of the generative models is usually very simple: unimodal and fully factorized (e.g., multivariate Gaussians or products of multinomials). For binary data, the decoding distribution can be substituted with a Bernoulli distribution:

$$p_\theta(\hat{x} \mid z) = \mathrm{Bernoulli}\left(\hat{x}; p_\theta(z)\right),$$

and the probability mass function (PMF) is defined as:

$$p_\theta(\hat{x} = k \mid z) = \begin{cases} p_\theta(z) & \text{if } k = 1, \\ 1 - p_\theta(z) & \text{if } k = 0. \end{cases}$$

During the training process, Stochastic Gradient Variational Bayes (SGVB)[37] estimator is employed via the reparameterization trick, and the parameters can be optimized via Auto-Encoding VB (AEVB)[37] algorithm.

Importance sampling

Importance sampling (IS)[34] is a technique used in statistical inference and Monte Carlo methods to estimate properties of a target distribution $p(x)$. Suppose direct sampling from $p(x)$ is not feasible or inefficient. We start by defining the proposal Probability Density Function (pdf), $q(x)$, from which we can easily generate the samples. Consider the following intractable integrals and the trick is the followings:

$$I = \int_{x \sim p(x)} f(x) dx = \int_{x \sim q(x)} \frac{p(x) f(x)}{q(x)} dx. \tag{1}$$

IS tries to improve this estimate by sampling x^n from proposal density q rather than p, the procedure consists of two simple steps:

1. Sampling: N samples are generated by

$$x^n \sim q(x), n = 1, \ldots, N.$$

2. Weighting: each sample is assigned an importance weight, which is determined by

$$w_n = \frac{p\left(x^n\right)}{q\left(x^n\right)}, n = 1, \ldots, N. \tag{2}$$

IS allows for an approximation of Eq. (1) through the self-normalized IS estimator,

$$\hat{\mathrm{I}} = \sum_{n=1}^{N} \bar{w}_n f\left(x^n\right),$$

where $\bar{w}_n = \frac{w_n}{\sum_{i=1}^{N} w_i}$ are the normalized weights. The IS methodology can be readily extended to cases where the samples are generated from multiple proposals instead of a single one. This extension is usually called multiple importance sampling (MIS). MIS schemes consider a collection of M proposal probability density functions, denoted as $\{q_n(x)\}_{n=1}^{M} \equiv \{q_1(x), \ldots, q_M(x)\}$, and proceed by drawing N samples, $\{x^m\}_{m=1}^{N}$ (where N may differ from M) and accurately weighting them.

Though there are many possible weighting schemas for the M proposals, we are only interested in the deterministic mixture multiple importance sampling (DM-MIS)[36], also known as the N3 scheme. This schema does not consider the prior information, and can be simply represented as a single mixture proposal with equal weights,

$$\psi(x) \equiv \frac{1}{M} \sum_{j=1}^{M} q_j(x),$$

then the weighting can be represented as

$$w_n = \frac{p\left(x^n\right)}{\psi\left(x^n\right)} = \frac{p\left(x^n\right)}{\frac{1}{M}\sum_{j=1}^{M} q_j\left(x^n\right)}, n = 1, \ldots, N.$$

This re-weighting allows for substantial variance reduction of the MIS estimator. The trade-off, however, is that the computational cost increases with the number of iterations, growing quadratically as the iterations progress.

Methods

In this section, we employ variational inference framework to formulate the problem of inferring missing data and present our proposed DLVM learning method.

Problem formulation

We consider the case where part of elements of the data is missing in the training data. Given the training data $\mathrm{X} = (x_1, \ldots, x_n)^\top \in \mathrm{R}^{n \times d}$, where the i-th data-point x^i is divided into two parts: observation x^i_{obs}, missing part x^i_{mis}, and $x^i = x^i_{\text{obs}} \cup x^i_{\text{mis}}$. The marginal likelihood of the incomplete data is typically written as

$$\ell(\boldsymbol{\theta}) = \log p_{\boldsymbol{\theta}}(\mathrm{X}) = \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}\left(x^i_{\text{obs}}\right), \tag{3}$$

where $p_\theta\left(x^i_{\text{obs}}\right) = \int p_\theta\left(x^i_{\text{obs}} \mid z\right) p(z) dz$. The goal is to find a MLE $\hat{\theta} \in \operatorname{argmax}_\theta \ell(\theta)$. However, $p_\theta\left(x^i_{\text{obs}}\right)$ is intractable so that $\ell(\theta)$ is intractable. Under the VAEs framework, we derive Eq. (3) as follows:

$$\ell(\theta) \geq \sum_{i=1}^{n} \mathbb{E}_{z^i \sim q_\phi}\left(z \mid x^i_{\text{obs}}\right)\left[\log \frac{p_\theta\left(x^i_{\text{obs}}, z^i\right)}{q_\phi\left(z^i \mid x^i_{\text{obs}}\right)}\right] = \mathscr{L}_{\text{ELBO}}.$$

Learning $\mathscr{L}_{\text{ELBO}}$ for incomplete data is similar to that for fully observed data. However, there are several distinctions, which are outlined below:

- Encoder $q_\phi\left(z \mid x_{\text{obs}}\right)$:

$$\begin{aligned} x &= 0 \odot (1-m) + x_{\text{obs}} \odot m, \\ (\mu, \sigma) &= \text{EncoderNeuralNet}_\phi(x), \\ q_\phi\left(z \mid x_{\text{obs}}\right) &= \mathscr{N}(z; \mu, \sigma), \end{aligned}$$

where $\odot$ is the Hadamard product. The encoder employs a deep neural network to generate the mean and variance. Subsequently, we model the encoder as a normal distribution to facilitate sampling of z using reparameterization techniques. Different from the traditional VAEs, we here use the incomplete data as input and the missing elements are imputed with 0 before training.

- Decoder $p_\theta\left(x_{\text{obs}} \mid z\right)$:

$$\begin{aligned} y &= \text{DecoderNeuralNet}_\theta(z), \\ \log p_\theta\left(x_{\text{obs}} \mid z\right) &= \log \text{Bernoulli}(x; y) \\ &= \sum_{j \in \Omega} x_j \log y_j + (1-x_j) \log\left(1-y_j\right). \end{aligned}$$

The decoder is to reconstruct the input data from the latent space representation z learned by the encoder, and transforms it back into the original input space via deep neural network. As we primarily focus on the missing data imputation of 2D data, such as images, we consider the Bernoulli distribution as the decoder. Under MAR assumption, the decoder incorporates both the observed and missing data distribution information. During training, only the observed data values are used to compute the reconstruction loss. However, after training, the decoder can be utilized to generate samples of the missing elements in the imputation step.

Fit VAEs with incomplete data via MISELBO

Since the target density cannot be easily characterized when comes with incomplete data, this means that finding a single good proposal pdf, $q(x)$, is not always possible. To ameliorate the rise in posterior complexity resulting from missing data and enhance the estimation accuracy, a robust alternative consists of using a set of proposal pdfs[35,36]. This enables us to employ distribution families for the mixture components that are acknowledged to perform effectively under complete data scenarios, and utilize these mixtures to address the augmented complexity in the posterior caused by missing data.

We adopt a more manageable approach and define the variational distribution $q_{\boldsymbol{\phi}}\left(z \mid x_{\text{obs}}\right)$ as an ensemble distribution:

$$q_\phi\left(z \mid x_{\text{obs}}\right) = \frac{1}{S} \sum_{s=1}^{S} q_{\boldsymbol{\phi}_s}\left(z \mid x_{\text{obs}}\right), \tag{4}$$

where each component distribution $q_{\boldsymbol{\phi}_s}\left(z \mid x_{\text{obs}}\right)$ is an encoder network. A variance analysis has been proposed in Ref.[36] demonstrated that incorporating the mixture components consistently yields estimators with lower variance compared to using single one. Recent work has shown that as the ELBO becomes tighter, the approximation aligns more closely with the true posterior distribution. Consequently, this leads to improved accuracy and reliability in inference results. By tightening the ELBO, we can enhance the quality of the variational approximation and effectively facilitate probabilistic predictions by learning the latent variables. The MIWAE bound extends the Importance Weighted Evidence Lower Bound (IWELBO), which corresponds to the case where $S=1$. This bound is specifically designed for handling missing data and is defined as follows:

$$\mathscr{L}_{\text{MIWAE}}^{L} = \sum_{i=1}^{n} \mathbb{E}_{\left\{z^\ell\right\}_{\ell=1}^{n} \sim q_\phi\left(z \mid x_{\text{obs}}^i\right)} \left[\log \frac{1}{L} \sum_{\ell=1}^{L} \frac{p_\theta\left(z^\ell, x_{\text{obs}}^i\right)}{q_\phi\left(z^\ell \mid x_{\text{obs}}^i\right)}\right]. \tag{5}$$

We have $\ell(\boldsymbol{\theta}) \geq \mathscr{L}_{\text{MIWAE}}^{L} \geq \mathscr{L}_{\text{ELBO}}$, equality between MIWAE and ELBO holds iff $L=1$. Compare with vanilla ELBO, this augmentation allows for more precise estimation and handling of missing values. By assigning higher weights to samples that offer better explanations of the observed data, the MIWAE bound helps mitigate potential bias caused by missing data and enhances imputation accuracy. Additionally, the MIWAE bound enables the model to multiple imputations, effectively capturing the uncertainty associated with missing values. Leveraging the MIWAE bound in missing data imputation yields more robust and reliable results, providing a principled and statistically rigorous approach to address the challenge of handling missing data. A study in Ref.[15] showed that MISELBO provides a tighter bound than the average of standard ELBOs, and empirical evidence

demonstrates that it yields tighter bounds than the average of MIWAE. In the case of missing data, MISELBO can be defined as:

$$\mathscr{L}^{L}_{\text{MISELBO}} = \sum_{i=1}^{n}\sum_{s=1}^{S}\frac{1}{S}\mathbb{E}_{\{z^{s,\ell}\}_{\ell=1}^{L}}\left[\log\frac{1}{L}\sum_{\ell=1}^{L}\frac{p_\theta\left(x^i_{\text{obs}}, z^{s,\ell}\right)}{\frac{1}{S}\sum_{s=1}^{S}q_{\phi_s}\left(z^{s,\ell}\mid x^i_{\text{obs}}\right)}\right],, \tag{6}$$

where $\left\{z^{s,\ell}\right\}_{\ell=1}^{n} \sim q_{\phi_s}\left(z \mid x^i_{\text{obs}}\right)$, L is the number of importance samples, S is the number of mixture component to average ensemble. When $S = 1$, the bounds $\mathscr{L}^{L}_{\text{MISELBO}}$ correspond to $\mathscr{L}^{L}_{\text{MIWAE}}$, which are widely used for imputing the missing data via VAEs. However, as $S > 1$, the proposed bounds have the potential to be tighter, thanks to the increased flexibility of the variational distribution $q_\phi\left(z \mid x_{\text{obs}}\right)$. The importance-weighted bounds maintain the asymptotic consistency guarantees and approach the true marginal log-likelihood $p_\theta\left(x_{\text{obs}}\right)$ as $S \cdot L \to \infty$. This feature allows for more accurate estimation, at the cost of an increasing computational budget. The procedure of learning VAEs with MISELBO is depicted in Fig. 1.

This ensemble consists of multiple encoders that share the same decoder. The results are ensembled by averaging the importance-weighted ELBO. This process is easy to implement via the reparameterization tricks, and the likelihood is tractable. While there have attempts to learn the mixture Gaussian for the ensemble distribution, e.g.[38,39], however, these approaches introduce a non-differentiable category distribution, making it generally infeasible to directly apply the reparameterization trick. According to the analysis of MISELBO with fully observed data, it is well-suited for handling incomplete data and yields comparable results, that is, $\mathscr{L}^{1}_{\text{MISELBO}} \leq \cdots \leq \mathscr{L}^{L-1}_{\text{MISELBO}} \leq \mathscr{L}^{L}_{\text{MISELBO}} \underset{L\to\infty}{\longrightarrow} \ell(\boldsymbol{\theta})$, where $\mathscr{L}^{1}_{\text{MISELBO}}$ is the average ELBO of all the encoders. The inequality implies that $\mathscr{L}^{L}_{\text{MISELBO}}$ serves as a lower bound for $\log p_\theta(\text{X})$, and is tighter than average ELBO. A larger L leads to a closer approximation to the true log-likelihood. The details of learning the ensemble VAEs for incomplete data is presented in Algorithm 1.

Input: Dataset $D = \left\{x^i_{obs}\right\}_{i=1}^{n}, S, L$, batch size: bs, learning rate: lr, max iteration: T.
Output: Model parameters $\Theta = \{\theta, \{\phi_s\}_{s=1}^{S}\}$
for t = 1 to T **do**
  Randomly divided dataset $\mathscr{D}$ into k ($k = \lfloor n/bs \rfloor$) batches $\mathscr{D} = \bigcup_{j=1}^{k}\mathscr{D}_j$;
  Set $\tilde{\mathscr{L}} \leftarrow 0$
  for j = 1 to k **do**
    (1) Compute $q_{\phi_s}\left(z \mid x^i_{\text{obs}}\right) := \mathscr{N}(z; \mu, \sigma)$ by the encoder neural network for all $x^i_{\text{obs}} \in \mathscr{D}_j, s = 1, .., S$
    (2) Sample $z^{s,\ell} \sim q_{\phi_s}\left(z \mid x^i_{\text{obs}}\right)$ for all $x^i \in \mathscr{D}_j, s = 1, ..S, \ell = 1, .., L$
    (3) Compute $p_\theta\left(x^i_{\text{obs}} \mid z^{s,\ell}\right)$ by the decoder neural network for all $x^i \in \mathscr{D}_j, s = 1, \ldots, S, \ell = 1, \ldots, L$
    (4) Compute $p\left(z^{s,\ell}\right)$ by standard Normal distribution for $s = 1, \ldots, S, \ell = 1, .., L$
    (5) $\tilde{\mathscr{L}} \leftarrow \tilde{\mathscr{L}} + \sum_{i\in\mathscr{D}_j}\frac{1}{S}\log\frac{1}{L}\sum_{\ell=1}^{L}\frac{p_\theta\left(x^i_{\text{obs}}|z^{s,\ell}\right)p\left(z^{s,\ell}\right)}{\frac{1}{S}\sum_{j=1}^{S}q_{\phi_j}\left(z^{s,\ell}|x^i_{\text{obs}}\right)}$
    (6) $\{\phi_s\}_{s=1}^{S}, \theta \leftarrow$ Back-propagation and stochastic gradient descent step for $\tilde{\mathscr{L}}$
    (7) Update model parameters Θ
  end for
end for

Algorithm 1. Learning the MMISVAE model.

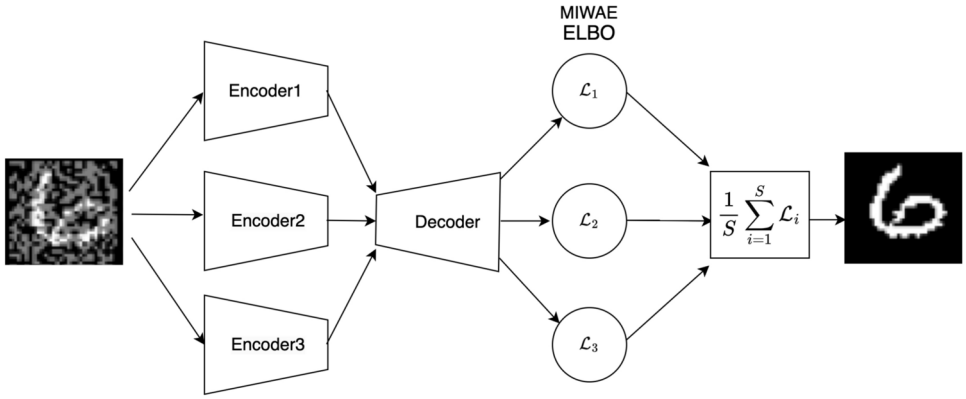

Fig. 1. The structure of the ensemble model, specifically focusing on the case where the number of mixture components is $S = 3$.

Imputation

Imputation with single proposal pdf

After learning the DLVM parameters θ and $\{\phi_s\}_{s=1}^{S}$, we can employ the model to infer the missing part of the data x_{mis} given the partial observation x_{obs} via the posterior $p_\theta(x_{\text{mis}} \mid x_{\text{obs}})$. The aim is to find a good imputation $\hat{x}_{\text{mis}}$ that is close to the true x_{mis}. The general optimal strategy would be using the conditional expectation, that is,

$$\begin{aligned}\hat{x}_{\text{mis}} = \mathbb{E}\left[x_{\text{mis}} \mid x_{\text{obs}}\right] &= \int x_{\text{mis}} p_\theta\left(x_{\text{mis}} \mid x_{\text{obs}}\right) dx_{\text{mis}} \\ &= \iint x_{\text{mis}} p_\theta\left(x_{\text{mis}} \mid x_{\text{obs}}, z\right) p_\theta\left(z \mid x_{\text{obs}}\right) dz dx_{\text{mis}}.\end{aligned} \tag{7}$$

Since $p_\theta(z \mid x_{\text{obs}})$ is intractable, by using the conditional probability formulas: $p_\theta(z \mid x_{\text{obs}}) = p_\theta(z, x_{\text{obs}}) / p_\theta(x_{\text{obs}})$, we rewrite (7) as

$$\begin{aligned}\hat{x}_{\text{mis}} &= \iint x_{\text{mis}} \frac{p_\theta\left(x_{\text{obs}}, z\right)}{q_\phi\left(z \mid x_{\text{obs}}\right) p_\theta\left(x_{\text{obs}}\right)} p_\theta\left(x_{\text{mis}} \mid x_{\text{obs}}, z\right) q_\phi\left(z \mid x_{\text{obs}}\right) dx_{\text{mis}} dz \\ &= \iint\limits_{(x_{\text{mis}}, z)} x_{\text{mis}} \frac{p_\theta\left(x_{\text{obs}}, z\right)}{q_\phi\left(z_{\text{obs}}\right) p_\theta\left(x_{\text{obs}}\right)} dx_{\text{mis}} dz,\end{aligned} \tag{8}$$

where $(x_{\text{mis}}, z) \sim p_\theta(x_{\text{mis}} \mid x_{\text{obs}}, z)\, q_\phi(z \mid x_{\text{obs}})$, and the single proposal distribution $p_\theta(x_{\text{mis}} \mid x_{\text{obs}}, z)\, q_\phi(z \mid x_{\text{obs}})$ is introduced. Then we can generate a set of L samples: $\left(x_{\text{mis}}^1, z^1\right), \ldots, \left(x_{\text{mis}}^L, z^L\right)$ through straightforward ancestral sampling from the proposal. Specifically, we first sample z from $q_\phi(z \mid x_{\text{obs}})$, then sample x from the decoder $p_\theta(x \mid z)$, after that, we obtain $x_{\text{mis}} = p_{\Omega^\perp}(x)$. Similar to (2), we define the weights as

$$w_l = \frac{p_\theta\left(x_{\text{obs}}, z^l\right)}{q_\phi\left(z^l \mid x_{\text{obs}}\right) p_\theta\left(x_{\text{obs}}\right)}, l = 1, \ldots, L,$$

then Eq. (8) can be approximated through the self-normalized IS estimator:

$$\hat{x}_{\text{mis}} \approx \sum_{l=1}^{L} \bar{w}_l x_{\text{mis}}^l,$$

where $\bar{w}_l = \frac{w_l}{\sum_{l=1}^{L} w_l}$. Note that as due to the self-normalization, we can eliminate the term $p_\theta(x_{\text{obs}})$ in $\bar{w}_l$, and also neglect this term in w_l in practice.

Imputation with multiple proposal pdfs

In contrast to a single proposal, we now explore the utilization of multiple proposals for imputation. Specifically, we employ the Multiple Importance Sampling (MIS) method, which involves making slight modifications to adapt it for missing data imputation. The fundamental concept behind MIS is to sample from the target distribution using multiple importance weights and subsequently assign weights to the generated samples based on their magnitudes, ensuring that samples with higher weights more accurately represent the target distribution. The procedure comprises four main fundamental steps:

- Sampling: generating samples from a set of proposals;
- Weighting: evaluating the normalized importance weight of each sample;
- Resampling: sampling $S \times L$ samples with replacement according to a multinomial distribution $P(X = z^{s,l}) = \bar{w}_l^s$ $(s = 1, \ldots, S, l = 1, \ldots, L)$;
- Imputation: sample imputation from the generative models.The specifics of a single iteration of the imputation process are outlined in Algorithm 2. Incorporating resampling-based multiple proposal methods allows effective selection of samples crucial for imputation from S proposal distributions, while discarding unimportant samples. By repeatedly sampling and re-weighting, the sample set can better represent the original data distribution and increase diversity among samples. It can also effectively reduces estimation bias caused by insufficient or unbalanced sample sizes, improving estimation accuracy and stability.

1. Draw L samples of from each of the S proposal pdfs:

$$z^{s,l} \sim q_{\phi_s}\left(z \mid x_{\text{obs}}\right), \quad l = 1,\ldots,L; s = 1,\ldots,S$$

2. Calculate the $L \times S$ importance weights:

$$w_l^s = \frac{p_\theta\left(x_{\text{obs}} \mid z^{s,l}\right) p\left(z^{s,l}\right)}{\frac{1}{S}\sum_{s=1}^{S} q_{\phi_s}\left(z^{s,l} \mid x_{\text{obs}}\right)}, \quad l = 1,\ldots,L; s = 1,\ldots,S$$

3. Normalize the importance weights:

$$\bar{w}_l^s = \frac{w_l^s}{\sum_{s=1}^{S}\sum_{l=1}^{L} w_l^s}, \quad l = 1,\ldots,L; s = 1,\ldots,S$$

4. Resample $L \times S$ samples of z from the proposed set:

$$\bar{z}^i \sim \text{Multinomial}\left(\left\{z^{s,l}, \bar{w}_s^l\right\}_{l=1,s=1}^{L,S}\right), \quad i = 1,\ldots,L \times S$$

5. Sample $L \times S$ imputation:

$$x_{\text{mis}}^i \sim p_\theta\left(x_{\text{mis}} | \bar{z}^i, x_{\text{obs}}\right), \quad i = 1,\ldots,L \times S$$

Algorithm 2. Multiple importance resampling imputation of a single iteration.

Through Algorithm 2, we can obtain the reconstructed missing data from the estimated imputation via conditional expectation:

$$\hat{x}_{\text{mis}} = \frac{\sum_{i=1}^{L \times S} x_{\text{mis}}^i \cdot \overline{w}(\bar{z}^i)}{\sum_{i=1}^{L \times S} \overline{w}(\bar{z}^i)}, \tag{9}$$

where $\overline{w}(\bar{z}^i)$ is the corresponding weights (probability) of $\bar{z}^i$, and finally, the complete reconstructed data from the estimated imputation is :

$$\hat{x} = \hat{x}_{\text{mis}} \odot (1 - m) + x_{\text{obs}} \odot m. \tag{10}$$

In certain cases, it can be beneficial to sample from the posterior distribution, and generate multiple imputations that provide valuable insights into the captured variance of the model. In Algorithm 2, steps 4 and 5 can be repeated multiple times (e.g., $M = 20$) to generate M complete imputed datasets, each containing distinct plausible values for the missing data. Inference is then performed by combining information from all plausible imputed results using Rubin's rules[3], i.e., compute an overall mean and variance estimate for a specific statistic.

Experiments

In this section, we investigate the proposed methods on the following three extensively researched datasets and assess the learning and imputation outcomes through visual and quantitative evaluations.

Datasets

- The MNIST dataset (http://yann.lecun.com/exdb/mnist) contains 60,000 training samples and 10,000 testing samples. Each sample is a 28 × 28 pixel grayscale image representing a handwritten digit (a number between 0 and 9). Each pixel in the image represents its grayscale level, with values ranging from 0 to 255.
- The images in the Omniglot dataset (https://github.com/brendenlake/omniglot) consists of handwritten characters from 50 different writing systems, including languages such as Greek, Korean, Hebrew, Arabic, and more. The dataset was created by collecting images of characters written by different individuals. One of the unique features of the Omniglot dataset is the large number of different character classes it contains. For each writing system, the dataset includes 20 examples of each character.
- The Fashion MNIST datasets (https://github.com/zalandoresearch/fashion-mnist) contains 70,000 grayscale images with 10 different categories, each image has a size of 28 × 28 pixels. These images represent various types of clothing and accessories, such as T-shirts, trousers, shirts, dresses, shoes, etc. The images are presented in grayscale, with pixel values ranging from 0 to 255. Compared to the two aforementioned handwritten digit datasets, Fashion MNIST comprises images of clothing items that exhibit notable visual similarities (e.g., "shirts" and "coats") and contain richer details and texture information. These characteristics render unsupervised imputation significantly more challenging.The images in the three datasets were binarized by initially scaling them to the range [0,1] and then rounding the values. This process resulted in binary images with only two values, indicating that each pixel could only assume either 0 or 1. In a binary image, 0 usually represents the background or the absence of an object, while 1 represents the foreground or the presence of an object. All images are cropped to a size of 28 × 28 with single channel. The MNIST and fashion MNIST dataset is randomly divided into three parts with 50,000 samples for the training set, 10,000 samples for the validation set, and 10,000 samples for the testing set. The Omniglot dataset employs the 'background' set,

which comprises 19,280 samples, as the training set, while the 'evaluation' set, consisting of 13,180 samples, serves as the testing set.

Parameters setting and evaluation metric

Our experimental hardware configuration includes an Intel i9 CPU, 32 GB of memory, and an Nvidia graphics card model 3090. The experiments were conducted using PyTorch 2.1 on Ubuntu 22.04. We consider both MAR and MCAR case. In MCAR, the datasets are with 50% of the pixels missing uniformly at random. In MAR, a contiguous image block is randomly omitted. The parameter initialization and network model parameters were primarily referenced from[9]. The encoder and decoder are similar with[16], i.e., multi-layer perceptrons with 2 hidden layers and ReLU activations. The dimensions of both hidden layers in the encoder and decoder networks are set to 300. Besides, an additional ReLU activation function is added to the last layer of the decoder. Since the DLVM method we propose is completely unsupervised, the labels of the dataset are not taken into consideration. We have noticed that using carefully designed CNNs as the encoder and decoder (e.g., NVAE[40]) may outperform MLP on image datasets. However, this will lead to an increase in the complexity of experimental comparisons. Our experiment specifically aims to validate the efficacy of our proposed approach, hence we employ a consistent utilization of simple MLP as both the encoder and decoder for all the compared DLVM methods.

We will individually train three separate datasets and evaluate the results of the learning and imputation process. All datasets will utilize the identical network structure, facilitating model generalizability by simply resizing the dataset to 28×28. We use the same hyper-parameters for all the datasets, which are uniformly configured with a batch size of 100, a learning rate of 0.0005, and a latent vector dimension of 40. Unless stated otherwise, the default values in training for the epoch are set as 1000, and find the best epoch to save the model for imputation. The importance samples is set to $L = 50$, and the ensemble component $S = 3$.

For the binary datasets performance evaluation, we use the accuracy (ACC) and F1 score to evaluate the reconstruction accuracy. Both of the criterions mainly measure the accuracy of the ground true X and imputed data $\hat{\mathrm{X}}$ of binary images, which are defined as:

$$\mathrm{ACC}(\mathrm{X}, \hat{\mathrm{X}}) = \frac{\sum_{(i,j)\in\Omega^{\perp}} 1\left(X_{ij} == \hat{X}_{ij}\right)}{N_{\mathrm{mis}}}, \tag{11}$$

$$\mathrm{F1}(\mathrm{X}, \hat{\mathrm{X}}) = \frac{2 * \sum_{(i,j)\in\Omega^{\perp}} 1\left(X_{ij} == \hat{X}_{ij}\right)}{2 * \sum_{(i,j)\in\Omega^{\perp}} 1\left(X_{ij} == \hat{X}_{ij}\right) + \sum_{(i,j)\in\Omega^{\perp}} 1\left(X_{ij} \neq \hat{X}_{ij}\right)}. \tag{12}$$

ACC and F1 have higher values indicating better performance. The F1 score is the harmonic mean of precision and recall. It reaches its optimal value of 1 and its worst value of 0.

Convergence and imputation performance analysis

The initial experiments aim to assess the convergence of the proposed methods. We compare our approach with VAEs and MIWAE. It should be noted that VAEs can be viewed as the special case where $L = 1$ and $S = 1$, and MIWAE is fixed with $S = 1$. In contrast, our method includes the optimal selection of integer values for both S and L. The training losses of these three methods on binary MNIST and Omniglot are illustrated in Fig. 2. Note that for fair of comparisons, all the loss are the average across all training samples. As we can see in both figures, our method converges more rapidly compared to the other two methods. The presence of a larger number of classes in the Omniglot dataset, coupled with a limited 20 samples per class, and the absence of label information utilization in our method, result in a noticeable deceleration in convergence speed when the epoch value is set to 20. Nevertheless, even within the initial 20 epochs, the loss function value experiences a rapid decrease for all methods.

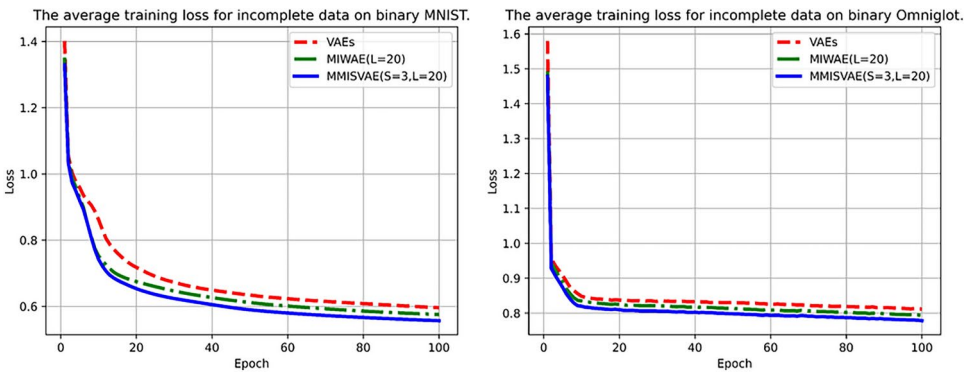

Fig. 2. The average training loss with different VAEs methods on incomplete on binary MNIST and Omniglot datasets.

Datasets	$S=1$	$S=3$	$S=5$	$S=7$
Binary MNIST	54.89	52.32	50.88	**50.38**
Fashion MNIST	126.84	124.02	122.15	**121.89**
Omniglot	79.75	77.02	76.81	**75.51**

Table 1. The MISELBO for an increasing number of mixtures (S) on different incomplete datasets. The missingness pattern is MCAR, with a 50% missing rate. Significant values are in bold.

Model	NLL
VAEs	55.32
MIWAE (L = 20)	54.19
MIWAE (L = 50)	53.38
MIWAE (L = 50) with multiple imputation	53.58
MMISVAE (S = 1, L = 50)	52.82
MMISVAE (S = 3, L = 50)	50.11
MMISVAE (S = 3, L = 50) with multiple imputation	**49.37**

Table 2. Testing NLL statistics for different VAEs methods on binary MNIST datasets. The missingness pattern is MCAR with 50% missing. Significant values are in bold.

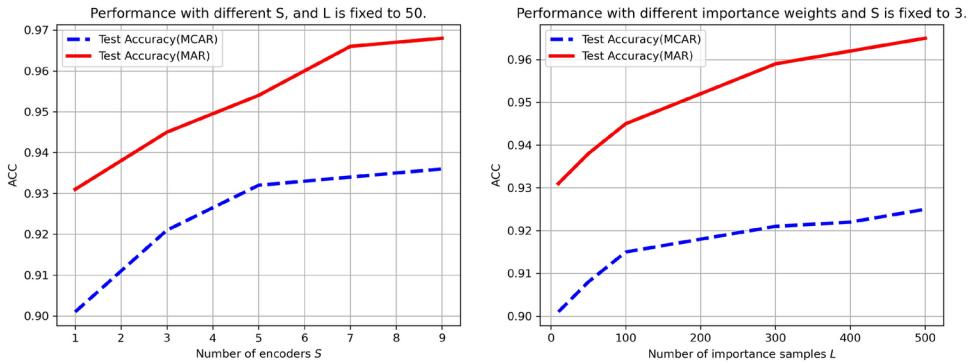

Fig. 3. Estimated test accuracy of the proposed method trained on binary MNIST (50% MCAR) with different S and L.

The second experiment primarily evaluates the performance of different mixture components. We set $L = 50$ and vary the values of S to analyze the training loss function. The results from different datasets are aggregated and presented in Table 1. The findings indicate a decrease in average loss function values as S increases. However, we should point out that the increase in S also results in a higher number of network parameters, leading to longer training times. Nevertheless, the overall time increase remains within an acceptable range. In the experiment, when S increased from 1 to 7, the training time will increase, but still within an acceptable range. In fact, the greater the value of S, the more pronounced the imputation effect. However, as S increases, the significance of the effect diminishes, resulting in a substantial increase in both training and testing time. Therefore, in general, opting for a smaller S is more suitable.

Next, we analyze the negative log-likelihood (NLL) of the test set for various methods. We record only the optimal values achieved. We compare VAEs using different parameters. To implement the multiple imputation approach, we generate 20 complete datasets and compute the average results. The NLL values for each method are presented in Table 2. The findings demonstrate improved performance when selecting larger and appropriate values for S and L. Furthermore, multiple imputation provides additional enhancements.

To evaluate the performance of ensembling multiple encoders and the impact of weighted importance sampling, we conducted two sets of experiments. In the first set, S was fixed at 3 while L was varied. In the second set, L was fixed at 50 while S was varied. The results, presented in Fig. 3, demonstrate that increasing the values of S and L improves testing accuracy. This provides empirical evidence supporting the effectiveness of our proposed multiple importance sampling method for missing data imputation.

Next, we consider the computational cost incurred by the increase in the number of encoders and importance samples. Here, we primarily analyze the time generated by different numbers of encoders during the training process, as well as the runtime caused by the increase in the number of importance samples during the imputation process. We consider the binary MNIST datasets and assume that the data has 50% random missing values

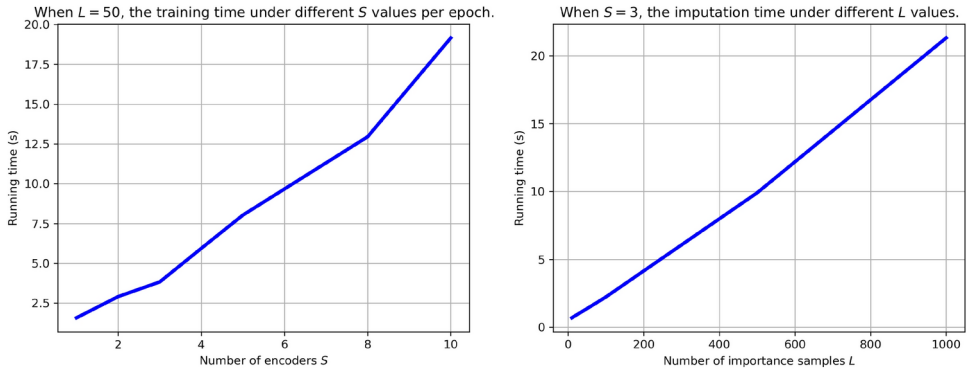

Fig. 4. The computational cost associated with different numbers of encoders S and importance samples L on binary MNIST (50% MCAR).

	MNIST		Omniglot		FashionMNIST	
Datasets	**ACC**	**F1 score**	**ACC**	**F1 score**	**ACC**	**F1 score**
Zero imputation	0.869	0.0	0.931	0.944	0.681	0.0
MissForest[41]	0.926	0.801	0.935	0.954	0.862	0.882
MICE[33]	0.935	0.803	0.921	0.933	0.871	0.883
GAIN[7]	0.915	0.788	0.892	0.902	0.832	0.826
MIWAE[9]	0.949	0.806	0.936	0.961	0.941	**0.909**
MMISVAE (ours)	**0.962**	**0.821**	**0.942**	**0.969**	**0.958**	0.909

Table 3. Test accuracy with different methods on different incomplete datasets (MCAR with 50% missing). Significant values are in bold.

(MCAR). The results are showed in Fig. 4. As can be observed from Fig. 4, an increase in the number of encoders leads to an approximately quadratic growth in training time, while an increase in the number of importance samples results in a linear growth in imputation time. Determining the optimal values for S and L primarily involves empirical trade-offs, which can be informed by integrating insights from Figs. 3 and 4. Moreover, since an increase in S results in a linear growth in network parameters, excessively large values may lead to overfitting. Nevertheless, to improve imputation performance, a larger L can be selected, provided sufficient computational resources are available. For instance, in practice, values of $L = 1000$ or higher are frequently selected.

Finally, the performance of different classic imputation methods on the three datasets is compared. Zero imputation is used as the baseline, where missing values are simply replaced with zeros. Given that the data values are binarized, zeros are the prevailing values, making zero imputation a reasonable approach. Additionally, a comparison is made between the classic methods MissForest, MICE, and the GAIN method based on GAN. MissForest and MICE are tested via the implementation of 'impute.IterativeImpute' class of Scikit-learn, and the GAIN is used the default parameters by the original authors. The testing accuracy and F1 scores for the different datasets are summarized in Table 3. The results demonstrate that our method surpasses others in terms of both accuracy and F1 score. Furthermore, both our method and the MIWAE method based on VAEs significantly outperform the other classic methods. Note that we compute the F1 score of the missing entries with Scikit-learn package, which is by default calculated as 0.0 when there are no true positives, false negatives, or false positives.

Data reconstruction

Below, we visualize the imputed missing pixels and reconstruct the image data. Our analysis begins by testing the Omniglot dataset, specifically focusing on the case of 50% missing pixels following the assumption of MCAR. The performance of VAEs and MIWAE is compared, and the results of the reconstruction are presented in Fig. 5 (left). While all three methods exhibit considerable performance, due to the relatively small number of samples and the large number of classes in this dataset, it is more challenging to train. We can see that they still lack some finer details when compared to the ground truth images. Notably, our method stands out by capturing more details, e.g., as evident in the reconstruction result of the last image, while VAEs display relatively poorer performance. The testing accuracy and the F1 score of these three methods are 0.921/0.913, 0.938/0.941, and 0.945/0.969, respectively.

Next, we evaluate the image imputation performance on the more challenging Fashion MNIST dataset using the same MCAR assumption and 50% data missing. We compare the performance of VAEs and MIWAE, and the results are presented in Fig. 5 (right). From the visualized results, all methods are capable of effectively imputing the missing data and generating decent reconstructed images overall. However, none of the methods are able to completely restore the ground truth images and have some limitations. The testing accuracy of the three

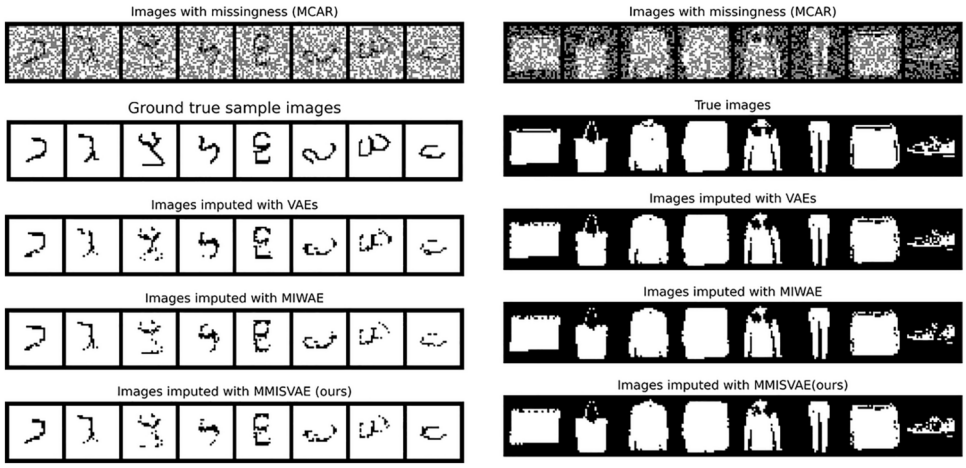

Fig. 5. Sample image reconstruction on Omniglot and FashionMNIST compared with different VAEs-based imputation methods.

Fig. 6. The sample image reconstruction with the proposed method was conducted on the binary MNIST, considering various mixture components and MAR conditions.

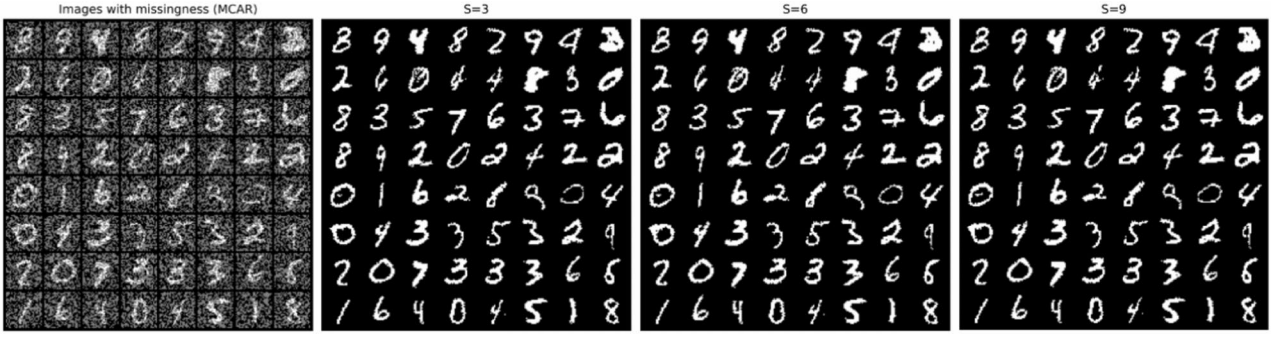

Fig. 7. The sample image reconstruction with the proposed method was conducted on the binary MNIST, considering various mixture components and MCAR conditions.

methods is as follows: 0.931, 0.941, and 0.952. Since we used a unified network architecture, further optimization of the network structure may be necessary to better adapt to this dataset.

Finally, we visually analyze the performance of reconstructed images under different components on the binary MNIST dataset. We fix $L = 50$ and vary the value of S. We assessed the imputation of missing data and image reconstruction under both MCAR and MAR scenarios, with the results displayed in Figs. 6 and 7. The best testing accuracies were 0.962 and 0.912 for MCAR and MAR cases, respectively. In both MAR and MCAR case, the best results are achieved at the largest S. However, for the MCAR case, there is little improvement among different components, with their ACC values being 0.951, 0.956, and 0.962, respectively. In contrast, for

the MAR case, the difficulty of imputation is significantly greater than that of MCAR. Therefore, increasing the value of component S noticeably improves the imputation effect. As we can see in Fig. 6, the reconstruction effect of $S = 5$ is significantly better than that of $S = 1$ in terms of visual appearance. Furthermore, we can observe in Fig. 7 that when the single encoder ($S = 1$) performs well, there is a modest improvement in utilizing mixture components ($S > 1$), but the improvement is not significant. However, when there is a noticeable discrepancy between the performance of a single encoder and the ground truth, using mixture components with multiple encoders can lead to a significant improvement. Through our experiments, we found that the advantages of Algorithm 2 lie in using multiple proposed distributions to generate various samples, and retaining samples crucial for imputing data through resampling, thereby maximizing the extraction of important samples generated by different encoders, and ultimately achieving stable imputation effects through conditional expectations.

Further discussions

In importance weighted sampling, the significance of samples is key for estimating conditional expectations. As our method involves sampling from the encoder (proposal distribution $q(z|x)$), the resulting samples may not precisely capture the characteristics of the true conditional distribution $p(z|x)$. To address this discrepancy, importance weights w_i are employed to correct the bias, ensuring that the contributions of the samples more accurately align with the true conditional distribution. Furthermore, these weights assign higher values to samples with greater probabilities under $p(z|x)$, thereby enhancing the fidelity of the conditional distribution representation. A well-constructed weight distribution can substantially reduce estimation variance and improve the precision of conditional expectation estimates. However, due to the inherent randomness of sampling, many sampled instances may be unsuitable for imputing missing values. This motivates our exploration of strategies to obtain higher-quality samples for data imputation. Our approach begins by generating a large, potentially redundant set of samples, computing their importance weights, and subsequently applying resampling techniques to identify samples better suited for missing value imputation. While both our current and prior work[42] are grounded in this concept, they diverge in their implementation methodologies, which we outline briefly below.

The key improvement of Ref.[42] over MIWAE lies in constructing the conditional distribution of missing data using observed data. This conditional distribution is modeled via beta-VAEs, and during the imputation phase, samples are iteratively drawn from this distribution, with each iteration generating new samples conditioned on previous generated samples. In contrast, the proposed method extends MIWAE by employing multiple encoders instead of a single one. It combines these encoders through an ensemble approach, samples from each encoder, and retains samples with higher weights. The resampling process is conducted among the samples generated by different encoders, ensuring a more robust selection of importance samples.

Both our method and prior approaches generate a significant number of samples through sampling and subsequently perform data imputation based on their importance weights. Previous methods iteratively produce samples, for instance, by iterating S times and generating L samples per iteration, resulting in a total of $S \times L$ samples. In contrast, the proposed method generates samples in a single iteration but employs multiple distinct encoders, e.g., by using S encoders, and each generating L samples, yielding the same total of $S \times L$ samples. Both strategies then perform imputation by weighting and combining these samples according to their importance weights. Hence, we believe that the implementation of more efficient sampling techniques for ensemble VAEs will further enhance the accuracy of the imputation performance.

Conclusions

In this study, we enhance the classical MIWAE method by employing multiple distinct encoders to ensemble the mixture components. Our work build upon on the recently proposed multiple importance sampling methods, which have demonstrated state-of-the-art results on complete datasets. Our method requires only minor modifications to be applied to missing data. The proposed method improves MIWAE in both model and imputation algorithm aspects. It generates multiple samples using different encoders and selects important samples through importance-weighted resampling, resulting in more stable and achieves better imputation performance than MIWAE. We conduct evaluations on several binary image datasets, and the results not only showcase the superiority of our method over current approaches but also validate its optimal performance compared to VAE-based imputation methods. However, since our method is constructed exclusively based on observed data, in the case of Missing Not At Random (MNAR), the missing mechanism depends on the unobserved data itself, rendering the assumptions underlying our conditional distribution invalid. Fortunately, recent studies have extended the MIWAE to MNAR case. For instance, NOT-MIWAE[29] constructs a missingness model to learn the mapping mechanism from data to the missing mask. Furthermore, Chen[43] proposes a deep generative imputation model that addresses real-world missing mechanisms in the latent space, simultaneously imputing incomplete data and reconstructing the missing data mask. The key insight from both approaches lies in the effective learning of the missing data mask. Consequently, leveraging the proposed ensemble VAEs to learn the missing data mask will be the focus of our future work, offering a potential solution to extend our method to the MNAR case.

Data availability

The dataset used in the experiment is publicly available and can be accessed through the following website: MNIST: http://yann.lecun.com/exdb/mnist; Omniglot: https://github.com/brendenlake/omniglot and FashionMNIST: https://github.com/zalandoresearch/fashion-mnist.

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Author contributions

S.K. wrote all paper sections and conducted most of the experiments. Y.H. co-proposed the idea and completed part of the experiments. J.S. provided feedback and corrections on all sections of the paper. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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